3:1

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1. 1
1.1. 1
1.2. 1
1.3. 2

2. - , 4
2.1. 3:1 - 4
2.2. 4
2.3. 5
2.4. 6
2.5. 8

3. 8
3.1. 8
3.2. 9
3.3. 9

4. 10
4.1. 10
4.1.1. 11
4.1.2. 13
4.2. 15
4.2.1. 15
4.2.2. 15
4.3. 17
4.3.1. 18

5. 23

6. 23
6.1. 1 23
6.2. 24

1.

1.1.

3:1 . .
: (S), (J) (A), , $m_A \ll m_J \ll m_S$. S J , A , SJ.
[1]:

$$H = \frac{p_x^2 + p_y^2}{2} + p_x y - p_y x + U(x, y, t),$$

$U(x, y, t)$ - , $r(t)$ S J.
$e_J > 0$ (0.05) . , . (S J , $r(t) = \text{const}$, $e_J = 0$), , (Wisdom, 1987, "Chaotic behavior in the Solar System Meyer, Hall, 2017, "Introduction to Hamiltonian Dynamical Systems and the N-Body Problem").

1.2.

$$\nu = \frac{m_J}{m_S + m_J} \approx 10^{-3},$$

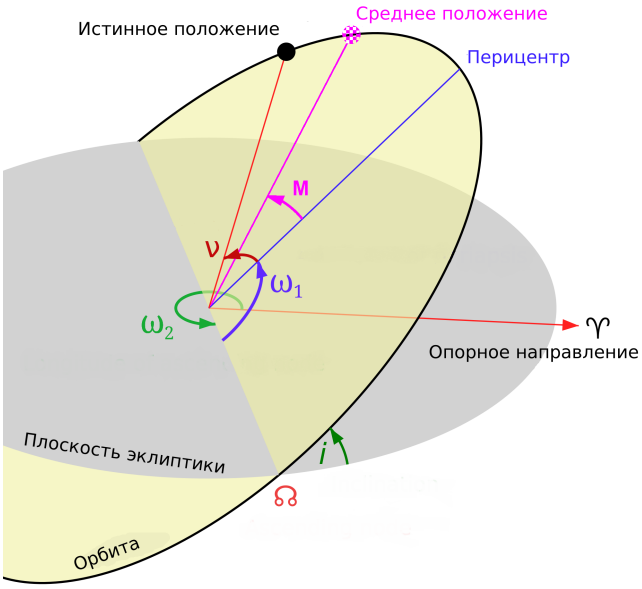
SJ.
(-) (MD, Plummer (1918) An Introductory Treatise on Dynamical Astronomy), [2]:

$$H = -\frac{(1-\nu)^2}{2L^2} - \nu R(L, \rho_1, \rho_2, l, \omega_1, \omega_2),$$

R :

$$R = \sum_{s-j-k+p-m-n=0,\ s\geq 0} K^{sjkpmn}(L, \rho_1, \rho_2) \cos(sl + pl_J + j\omega_1 + k\omega_2 + m\omega_{1J} + n\omega_{2J}).$$

L, ρ_1, ρ_2 : a, e i. – l, ω_1 ω_2. , , J.



. 1. $l = \omega_1 + \omega_2 + M$, M -

, , [3] ( ):

$$s\dot{l} + p\dot{l}_J + j\dot{\omega}_1 + k\dot{\omega}_2 + m\dot{\omega}_{1J} + n\dot{\omega}_{2J} \approx 0.$$

, l , :

$$s\dot{l} + p\dot{l}_J \approx 0 \qquad \Leftrightarrow \qquad \dot{l} \approx \frac{-p}{s}\dot{l}_J \equiv \frac{s+q}{s}\dot{l}_J.$$

q 3:1 $q = 2,\ s = 1.$
, ($i = 0,\ \rho_2 = 0,\ \omega_2 = 0$), e_J, [3]:

$$H = -\frac{(1-\nu)^2}{2L^2} - \nu R_{sec}(\rho_1, \omega_1) - \nu R_{res}(L, l, \rho_1, \omega_1),$$

$R_{sec}(\rho_1, \omega_1) = -2\rho_1 F - e_J G\sqrt{2\rho_1}\cos\omega_1$ - , ,
$R_{res}(L, l, \rho_1, \omega_1) = 2\rho_1 C\cos(l - \omega_1 - 3l_J) + e_J^2 E\cos(l - 3l_J)$ - , .
[3]:

$$a_J = 1, \dot{l}_J = 1.$$

C, D, E, F, G $a_{res} = ((1-\nu)/9)^{1/3}$ $\nu = 1/1047.355$ [4]:

$$C \approx 0.8631579, D \approx -2.656407, E \approx 0.3629536, F \approx -0.2050694, G \approx 0.1987054$$

:

$$\Lambda = L - L_{res}, \quad \lambda = l - 3l_J, \quad x = \sqrt{2\rho_1}\cos\omega_1, \quad y = \sqrt{2\rho_1}\sin\omega_1,$$

Λ, :

$$H = \alpha\frac{\Lambda^2}{2} + \nu\left(C(x^2 - y^2) + e_J Dx + e_J^2 E\right)\cos\lambda + \nu(2Cxy + e_J Dy)\sin\lambda + \nu e_J Gx + \nu F(x^2 + y^2), \quad (1)$$

$L_{res} = \left(\frac{(1-\nu)^2}{3}\right)^{\frac{1}{3}}$ - , :

$$\dot{\lambda} = \frac{\partial H}{\partial L}|_{L=L_{res}} = 0, \qquad \alpha = -\frac{3(1-\nu)^2}{L_{res}^4} < 0.$$

, y λ , x L . , λ, , :

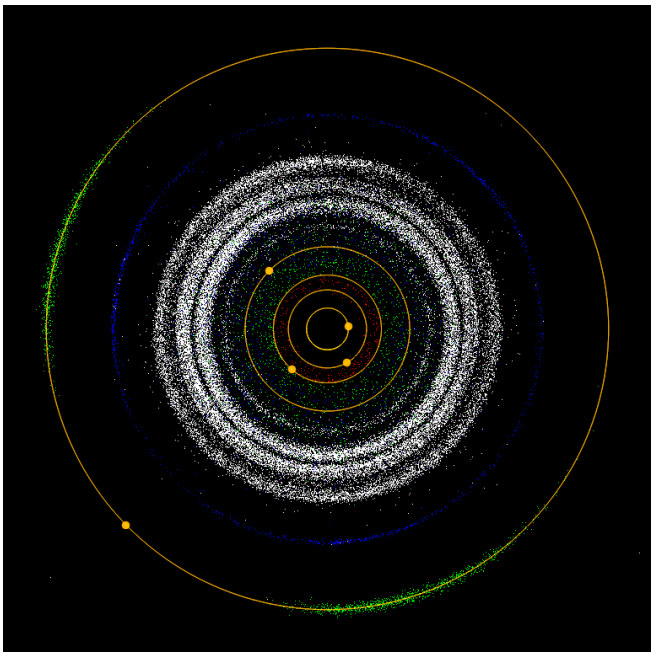
$$(x, y, \Lambda, \lambda) \in \mathbb{R}^3 \times S^1.$$

, (1), 3:1.

1.3.

– , . 1866 . . , . : 2:1, 3:1, 5:2, 7:3.

. 2.



. 3.

. , . [4]:

$$\begin{cases} \dfrac{d\Lambda}{dt} = \nu(U\sin\lambda - V\cos\lambda), & \dfrac{d\lambda}{dt} = \alpha\Lambda, \\ \dfrac{dx}{dt} = -\nu\Big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\Big), & \dfrac{dx}{dt} = \nu\Big(2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda\Big). \end{cases} \tag{2}$$

$$U(x,y) = C(x^2 - y^2) + e_J Dx + e_J^2 E, \qquad V(x,y) = 2Cxy + e_J Dy.$$

$\frac{dx}{dt}$ $\frac{dy}{dt}$ ν, x y ν^{-1}, λ ($\nu^{-\frac{1}{2}}$). , x y - : $x = \overline{x} + \xi$, $y = \overline{y} + \eta$, :

$$\begin{cases} \frac{d\overline{x}}{dt} = -\nu\left(2F\overline{y} + \frac{\partial U(\overline{x},\overline{y})}{\partial \overline{y}} < \cos\lambda > + \frac{\partial V(\overline{x},\overline{y})}{\partial \overline{y}} < \sin\lambda >\right), \\ \frac{d\overline{y}}{dt} = \nu\left(2F\overline{x} + e_J G + \frac{\partial U(\overline{x},\overline{y})}{\partial \overline{x}} < \cos\lambda > + \frac{\partial V(\overline{x},\overline{y})}{\partial \overline{x}} < \sin\lambda >\right), \end{cases} \tag{3}$$

T:

$$< \cos\lambda >= \frac{1}{T}\int_0^T \cos\lambda dt = \begin{cases} \frac{2E(k_L)}{K(k_L)} - 1, & -\sqrt{U^2+V^2} < H < \sqrt{U^2+V^2}, \\ \frac{2E(k_C)}{k_C^2 K(k_C)} + 1 - \frac{2}{k_C^2}, & H < -\sqrt{U^2+V^2}, \end{cases}$$

$$< \sin\lambda >= \frac{1}{T}\int_0^T \sin\lambda dt = 0,$$

$$k_L = \sqrt{\frac{\nu\sqrt{U^2+V^2} - H}{2\nu\sqrt{U^2+V^2}}}, \quad k_C = \sqrt{\frac{2\nu\sqrt{U^2+V^2}}{\nu\sqrt{U^2+V^2} - H}},$$

$K(k)$ - 1 , $E(k)$ - 2 .
1 , - .
, . , , λ $\overline{x}$ $\overline{y}$. $\frac{d\lambda}{dt} = O(\nu)$, , $\frac{dx}{dt}$ $\frac{dx}{dt}$ ν. , $U(x,y)\sin\lambda - V(x,y)\cos\lambda$ () .

2. - ,

2.1. 3:1 -

, (1) - . $\Lambda = \sqrt{\nu}\Lambda_{\text{new}}$, $t_{\text{new}} = \sqrt{\nu}t$. (“new”) $\varepsilon = \sqrt{\nu}$, (t):

$$\begin{cases} \dot{\Lambda} = U\sin\lambda - V\cos\lambda, & \dot{\lambda} = \alpha\Lambda, \\ \dot{x} = -\varepsilon\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & \dot{y} = \varepsilon\big(2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda\big). \end{cases} \tag{4}$$

λ Λ , x y .
, (4)

$$\Omega = d\Lambda \wedge d\lambda + \varepsilon^{-1} dx \wedge dy. \tag{5}$$

$$H = \alpha\frac{\Lambda^2}{2} + U(x,y)\cos\lambda + V(x,y)\sin\lambda + e_J G x + F(x^2+y^2), \tag{6}$$

1. , C, D, E, F, G :

$$C > |F|, \qquad D^2 - 4CE > 0$$
$$(DF - GC)^2 < (D^2 - 4CE)(\pm F - C)^2$$

2.2.

(4) :

$$M_0 = \{(x,y,\lambda,\Lambda): \quad \Lambda = 0, \quad u(x,y)\sin\lambda = v(x,y)\cos\lambda\}.$$

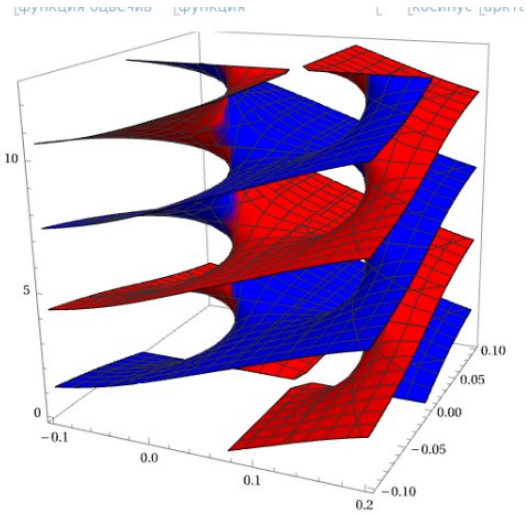
$U\sin\lambda = V\cos\lambda$ $\lambda = \arctan(V/U) + \pi k$, $k \in \mathbb{Z}$. $U = 0$ $\lambda_\pm : \mathbb{R}^2 \setminus \{U = V = 0\} \to S^1$, π:

$$\lambda_+(x,y) \equiv \begin{cases} \arctan\frac{V}{U}, & U > 0, V > 0, \\ \arctan\frac{V}{U} + \pi, & U < 0, \\ \arctan\frac{V}{U} + 2\pi, & U > 0, V < 0, \end{cases} \qquad \lambda_-(x,y) \equiv \begin{cases} \arctan\frac{V}{U} + \pi, & U > 0, V > 0, \\ \arctan\frac{V}{U} + 2\pi, & U < 0, \\ \arctan\frac{V}{U} + 3\pi, & U > 0, V < 0. \end{cases}$$

, $U = V = 0$, :

$$y = 0, \quad x = x_b^\pm \equiv \frac{-e_J D \pm e_J\sqrt{D^2 - 4EC}}{2C}, \quad \Lambda = 0, \quad \lambda \in S^1,$$

$\mathbb{R}^3 \times S^1$.
, , , $\tan\lambda = y/(x - x_b^\pm)$.



. 4. M_0 . $M_{0,+}$ – , $M_{0,-}$ –

, $M_{0,\pm}$, $M_{0,b}^{\pm}$:

$$
\begin{aligned}
M_0 &= M_{0,+} \cup M_{0,-} \cup M_{0,b}^{+} \cup M_{0,b}^{-},\\
M_{0,\pm} &= \{\lambda = \lambda_{\pm}(x,y), \Lambda = 0, (x,y) \in \mathbb{R}^2 \setminus \{U = V = 0\}\},\\
M_{0,b}^{\pm} &= \{y = 0, x = x_b^{\pm}, \Lambda = 0, \lambda \in S^1\}.
\end{aligned}
$$

$M_{0,\pm}$, $M_{0,b}^{\pm}$ - .

2.3.

(4) ε , "" :

$$
\begin{cases}
\dot{\Lambda} = U(x,y)\sin\lambda - V(x,y)\cos\lambda, & \dot{\lambda} = \alpha\Lambda,\\
\dot{x} = 0, & \dot{y} = 0.
\end{cases}
\tag{7}
$$

(x,y) U, V . λ t $\dot{\Lambda}$, :

$$\ddot{\lambda} - \alpha U \sin\lambda + \alpha V \cos\lambda = 0,$$

($\alpha < 0$) :

$$\ddot{\lambda} - \alpha\beta \sin\big(\lambda - \lambda_+(x,y)\big) = 0, \quad \beta(x,y) \equiv \sqrt{U^2 + V^2} > 0.$$

, ( ) "" , $(\lambda, \Lambda) = (\lambda_{\pm}(x,y), 0)$. (7) :

$$\begin{pmatrix} U\sin\lambda - V\cos\lambda \\ \alpha\Lambda \end{pmatrix} = \begin{pmatrix} 0 & \alpha \\ \pm\beta(x,y) & 0 \end{pmatrix} \begin{pmatrix} \lambda - \lambda_{\pm} \\ \Lambda \end{pmatrix} + O((\lambda - \lambda_{\pm})^2, \Lambda^2).$$

:

$M_{0,+}$:

$$\zeta = \pm i\sqrt{|\alpha|\beta},$$

$M_{0,-}$:

$$\zeta = \pm\sqrt{|\alpha|\beta}.$$

, - .
, $M_{0,+}$, $M_{0,-}$ - . , :

$$\lambda(t,t_0) = \pm 2 \arctan\sinh\big(\alpha\beta(t-t_0)\big) + \lambda_-(x,y), \qquad \Lambda(t,t_0) = \frac{\pm 2\beta}{\cosh\big(\alpha\beta(t-t_0)\big)}, \tag{8}$$

$t_0 \in \mathbb{R}$, .

2.4.

$\tau = \varepsilon t$ (4), (τ):

$$\begin{cases} \varepsilon\Lambda' = U\sin\lambda - V\cos\lambda, & \varepsilon\lambda' = \alpha\Lambda, \\ x' = -\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & y' = 2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda. \end{cases} \tag{9}$$

ε , "" :

$$\begin{cases} 0 = U\sin\lambda - V\cos\lambda, & 0 = \alpha\Lambda, \\ x' = -\big(2Fy + \dfrac{\partial U}{\partial y}\cos\lambda + \dfrac{\partial V}{\partial y}\sin\lambda\big), & y' = 2Fx + e_J G + \dfrac{\partial U}{\partial x}\cos\lambda + \dfrac{\partial V}{\partial x}\sin\lambda. \end{cases} \tag{10}$$

(10) , :

$$x' = -\big(2Fy \pm \frac{\partial\beta}{\partial y}\big), \quad y' = 2Fx + e_J G \pm \frac{\partial\beta}{\partial x}.$$

( ) $M_{0,+}$, $M_{0,-}$.
:

$$H_S = F(x^2+y^2) + e_J G x \pm \beta(x,y),$$

, .

$$(x_0^\pm, 0), \qquad x_0^\pm = -\frac{e_J(G \mp D)}{2(F \mp C)}.$$

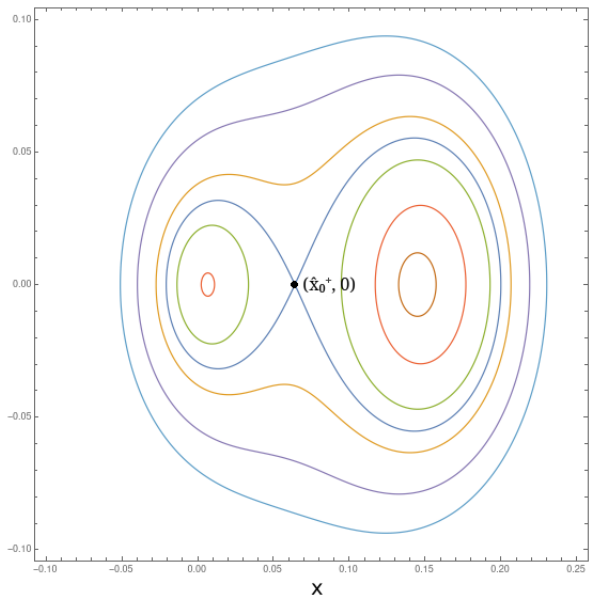
, , $+\xi_0^\pm, -\xi_0^\pm$ (M_0),

$$\xi_0^\pm = \sqrt{4(C^2-F^2) + 2\frac{(2C\hat{x}_0^\pm + e_J D)^2}{U(\hat{x}_0^\pm, 0)}(\pm F - C)} \in \mathbb{R}, \tag{11}$$

, 1, .
, () .
, ( ) , , $\tau \to \pm\infty$. , :

$$H_S(x,y) = H_S(x_0^\pm, 0),$$

$$H_S(x_0^\pm, 0) = F\big((x_0^\pm)^2\big) + e_J G x_0^\pm \pm \beta(x_0^\pm, 0).$$



. 5. H_S $M_{0,+}$. $M_{0,-}$

:

$$A = 2U(\hat{x}_0^{\pm})(\pm F - C), \qquad B = (2C\hat{x}_0^{\pm} + e_J D)(\pm F - C), \qquad J = (2C\hat{x}_0^{\pm} + e_J D) - 2U(\hat{x}_0^{\pm})(\pm F + C).$$

(X, Y):

$$x = \hat{x}_0^{\pm} + \frac{X - B/A}{(X - B/A)^2 + Y^2}, \qquad y = \frac{Y}{(X - B/A)^2 + Y^2}$$

:

$$AX^2 - JY^2 = (\xi_0^{\pm})^2/4.$$

1, $f \in \mathbb{R}$:

$$X = \frac{\cosh f}{s_1}, \qquad Y = \frac{\sinh f}{s_2}, \qquad s_1 \equiv \sqrt{4A/(\xi_0^{\pm})^2} \in \mathbb{R}, \qquad s_2 \equiv \sqrt{4J/(\xi_0 \pm)^2} \in \mathbb{R}.$$

f τ. $(x, y) \mapsto (X, Y)$,

$$f' = \frac{\pm|U(\hat{x}_0^{\pm})|\xi_0^{\pm}}{\beta(x(f), y(f))} = \pm|U(\hat{x}_0^{\pm})|\xi_0^{\pm}\frac{P(\cosh f)}{S(\cosh f)}$$

$$S(s) = -U_0 P(s) \mp F - (2C\hat{x}_0^{\pm} + e_J D)\left(\frac{s}{s_1} - \frac{B}{A}\right),$$

$$P = \frac{s_2^2 + s_1^2}{s_2^2}\left(\left(\frac{s}{s_1}\right)^2 - s\frac{2B}{As_1}\frac{s_2^2}{s_1^2 + s_2^2} + \left(\frac{B^2}{A^2} - 1/s_2^2\right)\frac{s_2^2}{s_1^2 + s_2^2}\right).$$

, f:

$$\tau - \tau_0 = \frac{\pm 1}{|U(\hat{x}_0^{\pm})|\xi_0^{\pm}}\int_{f_0}^{f}\frac{S(\cosh f)}{P(\cosh f)}df = \tag{12}$$

$$\frac{\pm 1}{\xi_0^{\pm}}(f - f_0) \pm \frac{s_1 s_2^2(2C\hat{x}_0^{\pm} + e_J D)}{|U(\hat{x}_0^{\pm}, 0)|\xi_0^{\pm}(s_1^2 + s_2^2)(X_2 - X_1)}\left(\left(X_1 - \left(\frac{B}{A} \mp \frac{F}{2C\hat{x}_0^{\pm} + e_J D}\right)\right) I(f, X_1 s_1) -\right.$$

$$-\left(X_2-\left(\frac{B}{A}\mp\frac{F}{2C\hat{x}_0^{\pm}+e_JD}\right)\right)I(f,X_2s_1)\Bigg),$$

$$I(f,a)\equiv\int_{f_0}^{f}\frac{dx}{\cosh x-a}=\frac{2}{\sqrt{1-a^2}}\arctan\sqrt{\frac{1+a}{1-a}}\tanh\frac{x}{2}\Big|_{f_0}^{f}.$$

X_1, X_2 P:

$$X_{1,2}=\frac{Bs_2^2}{A(s_1^2+s_2^2)}\pm\frac{2s_2^2(\pm F-C)}{(s_1^2+s_2^2)\xi_0^{\pm}}\sqrt{\frac{-C}{U_0}}$$

, $M_{0,+}$ $M_{0,-}$ (2.4) () $(x_0^{\pm},0)$, :

$$x(\tau,\tau_0)=x_0^{\pm}+\frac{\frac{\cosh(f(\tau-\tau_0))}{\sigma_1}-\frac{B}{A}}{P(\cosh(f(\tau-\tau_0)))},\quad y(\tau,\tau_0)=\frac{\sinh(f(\tau-\tau_0))}{\sigma_2P(\cosh(f(\tau-\tau_0)))},\quad \tau_0\in\mathbb{R}.$$

$f(\tau)$ (12).

2.5.

, $(\hat{x}_0^+,0,\lambda_+(\hat{x}_0^+,0),0)$, $(\hat{x}_0^-,0,\lambda_-(\hat{x}_0^-,0),0)$ (..), . ($\pm$):

$$\pm\frac{\sqrt{q\pm\sqrt{p}}}{\sqrt{2}},$$

(M_0)

$$p=\left(4\varepsilon^2(F^2-C^2)\mp|\alpha||U(\hat{x}_0^{\pm},0)|\right)^2+8\alpha\varepsilon^2(F\mp C)(2C\hat{x}_0^{\pm}+e_JD),$$
$$q=-4\varepsilon^2(F^2-C^2)\mp|\alpha||U(\hat{x}_0^{\pm},0)|.$$

1 :

- $(\hat{x}_0^-,0,\lambda_-(\hat{x}_0^-,0),0)\in M_{0,-}$ – - (, ,)
- $(\hat{x}_0^+,0,\lambda_+(\hat{x}_0^+,0),0)\in M_{0,+}$ – - $(\pm i\omega,\pm\xi)$.

$$\omega=\frac{\sqrt{\sqrt{p}-q}}{\sqrt{2}}=\sqrt{|\alpha U(\hat{x}_0^+,0)|}+\mathcal{O}(\varepsilon^2),$$
$$\xi=\frac{\sqrt{\sqrt{p}+q}}{\sqrt{2}}=\mathcal{O}(\varepsilon),$$

3.

3.1.

, . , [5].

:

$$\begin{cases}\frac{dX}{dt}=JD_XH_1(X,I)+\varepsilon g^X(X,I),\\ \frac{dI}{dt}=\varepsilon g^I(X,I),\end{cases}\tag{13}$$

$X\in\mathbb{R}^2$, $I\in\mathbb{R}^m$, D_X, D_I - , J - .
[5], $V\subset M_0$ - M_0 (13), V_ε - ε- V, . $\Gamma=W^s(V)\cap W^u(V)\setminus V$.

$W^s(V_\varepsilon)$ $W^u(V_\varepsilon)$ $p \in \Gamma$:

$$d^I(p, t_0, \varepsilon) = \varepsilon \frac{M^I(p, t_0)}{||D_X H(X_0^I, I)||} + O(\varepsilon^2), \tag{14}$$

M^I - , :

$$M^I(p, t_0) = \int_{-\infty}^{\infty} \Big(\langle D_X H_1, g^X \rangle + \langle D_I H_1, g^I \rangle \Big) \big(X_0^I(t - t_0, I), I \big) dt -$$
$$- \langle D_I H_1 \big(v(I), I \big), \int_{-\infty}^{\infty} g^I \big(X_0^I(t - t_0, I), I \big) dt \rangle.$$

$X = v(I)$ - V, $\langle \cdot, \cdot \rangle$ - $\mathbb{R}^2$.
$W^s(V_\varepsilon)$ $W^u(V_\varepsilon)$ $\varepsilon > 0$:

$$M^I = 0, \quad \nabla M^I \neq 0. \tag{15}$$

, , ε d^I .
.

3.2.

, M_0 $M_{0,+}$, $M_{0,-}$ $M_{0,b}^{\pm}$, $M_{0,+}$, $M_{0,-}$.
, $M_{0,-}$.
, (10) $M_{0,-}$ $(x(t), y(t), 0, \lambda_-(x(t), y(t)), \Lambda = 0)$ xy .
(4) (13) :

$$X = \begin{pmatrix} \Lambda \\ z \end{pmatrix}, \qquad I = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad z = \lambda - \lambda_-(x, y).$$

(4) (13):

$$D_X = \begin{pmatrix} \frac{\partial}{\partial \Lambda} \\ \frac{\partial}{\partial z} \end{pmatrix}, \qquad D_I = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}, \qquad H_1 = \alpha \frac{\Lambda^2}{2} + \beta(x, y)(1 - \cos z),$$

$$g^I = \begin{pmatrix} -2Fy + \frac{\partial \beta}{\partial y} \cos z + \beta(x, y) \frac{\partial}{\partial y} \arctan\left(\frac{V}{U}\right) \sin z \\ 2Fx + e_J G - \frac{\partial \beta}{\partial x} \cos z - \beta(x, y) \frac{\partial}{\partial x} \arctan\left(\frac{V}{U}\right) \sin z \end{pmatrix}, \qquad g^X = \frac{-1}{\varepsilon} \begin{pmatrix} 0 \\ \frac{d}{dt} \arctan\left(\frac{V}{U}\right) \end{pmatrix}.$$

3.3.

$V = M_{0,-}$:

$$M^I(x, y) = \frac{4}{\alpha \beta^2} \left(U(x, y) \left((2Fx + e_J G) \frac{\partial U}{\partial y} - 2Fy \frac{\partial U}{\partial x} \right) + V(x, y) \left((2Fx + e_J G) \frac{\partial V}{\partial y} - 2Fy \frac{\partial V}{\partial x} \right) \right). \tag{16}$$

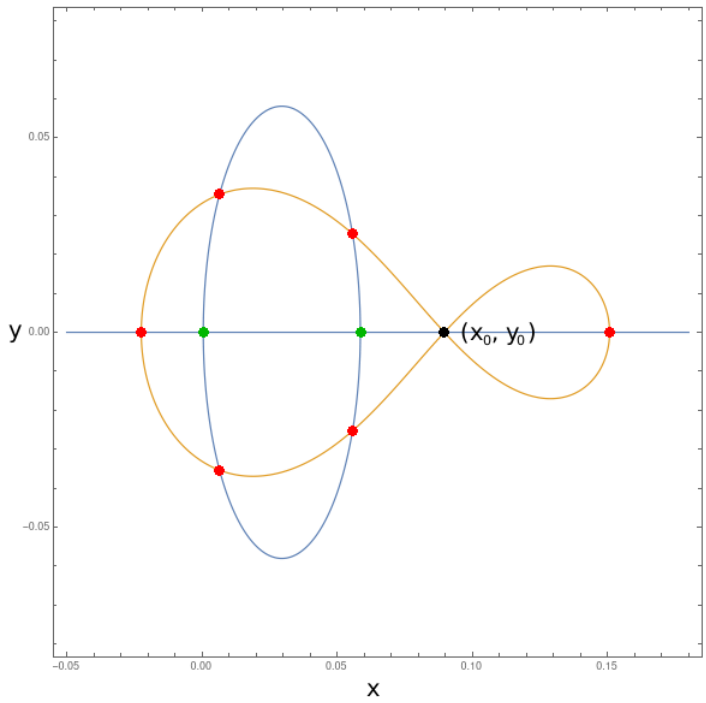
:

$$0 = y \left((2Cx + e_J D)^2 + C^2 y^2 + (2Cx + e_J D) \frac{F(D^2 - 4CE)}{GC - FD} + \frac{e_J^2 (D^2 - 4CE)}{4} \right), \tag{17}$$

$y = 0$ (. 6).
, , .
:

1) $M_{0,-}$, , $\varepsilon > 0$ ( ).

2) $\left(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)\right)$ $\varepsilon > 0$, . $\left(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)\right)$, (. 6).
, , $M_{0,-}$,



. 6. () (). – M^I

4.

.
, , , $(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0) \in M_{0,+}$, -, , , . , , . ε, , , .

4.1.

. , , : , .
- 0, :

$$\Lambda \to \Lambda, \qquad \lambda \to \lambda - \lambda_+(\hat{x}_0, 0) = \lambda - \pi, \qquad x \to x - \hat{x}_0, \qquad y \to y.$$

:

$$\begin{cases} \dot{\Lambda} = -U \sin\lambda + V \cos\lambda \equiv g^\Lambda(\Lambda, \lambda, x, y), , & \dot{\lambda} = \alpha\Lambda \equiv g^\lambda(\Lambda, \lambda, x, y), \\ \dot{x} = -\varepsilon\left(2Fy - \dfrac{\partial U}{\partial y}\cos\lambda - \dfrac{\partial V}{\partial y}\sin\lambda\right) \equiv g^x(\Lambda, \lambda, x, y), & \dot{y} = \varepsilon\left(2F(x + \hat{x}_0) + e_J G - \dfrac{\partial U}{\partial x}\cos\lambda - \dfrac{\partial V}{\partial x}\sin\lambda\right) \equiv g^y(\Lambda, \lambda, x, y). \end{cases} \tag{18}$$

$$U(x, y) = U_0 + x(2C\hat{x}_0 + e_J D) + C(x^2 - y^2), \qquad V(x, y) = 2Cxy + y(2C\hat{x}_0 + e_J D),$$
$$U_0 = U(\hat{x}_0^+, 0).$$

, (18) (5) :

$$H = \frac{\alpha\Lambda^2}{2} - U(x, y)\cos\lambda - V(x, y)\sin\lambda + F\left((x + \hat{x}_0)^2 + y^2\right) + e_J G x. \tag{19}$$

4.1.1.

(18), $W^{s,u}(0)$. (18) :

- $W^s(0)$:
$$\Lambda^s(+\infty)=0,\quad \lambda^s(+\infty)=0,\quad x^s(+\infty)=0,\quad y^s(+\infty)=0,$$
- $W^u(0)$:
$$\Lambda^u(-\infty)=0,\quad \lambda^u(-\infty)=0,\quad x^u(-\infty)=0,\quad y^u(-\infty)=0.$$

(18) :

$$\Lambda^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k\Lambda_k^{s,u}(\tau),\quad \lambda^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k\lambda_k^{s,u}(\tau),\quad x^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k x_k^{s,u}(\tau),\quad y^{s,u}(\tau)=\sum_{k=0}^{\infty}\varepsilon^k y_k^{s,u}(\tau),\tag{20}$$

, $\tau=\varepsilon t.$ ε, , , 2 , $(\lambda_k,\Lambda_k), k\geq 1$:

$$\begin{cases}\Lambda_k(\tau)=\frac{1}{\alpha}\lambda'_{k-1},\\ \lambda_k(\tau)=\frac{1}{\beta}\Big(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda}+x_k\big((2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0-2Cy_0\cos\lambda_0\big)+\\ \qquad\qquad +y_k\big(-2Cy_0\sin\lambda_0-(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big)\Big),\\ \lambda_{-1}=0,\end{cases}\tag{21}$$

$(x_k,y_k), k\geq 1$ - :

$$\frac{d}{d\tau}\begin{pmatrix}x_k\\ y_k\end{pmatrix}=\begin{pmatrix}a(\tau) & b(\tau)\\ c(\tau) & -a(\tau)\end{pmatrix}\begin{pmatrix}x_k\\ y_k\end{pmatrix}+\begin{pmatrix}\tilde{G}_k^x(\tau)\\ \tilde{G}_k^y(\tau)\end{pmatrix}\equiv\mathcal{A}(\tau)\begin{pmatrix}x_k\\ y_k\end{pmatrix}+\begin{pmatrix}\tilde{G}_k^x(\tau)\\ \tilde{G}_k^y(\tau)\end{pmatrix},\tag{22}$$

:

$$\tilde{G}_k^x=G_k^x-\beta^{-1}(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda})\big(-2Cy_0\sin\lambda_0-(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big),$$
$$\tilde{G}_k^y=G_k^y-\beta^{-1}(\alpha^{-1}\lambda''_{k-2}-G_k^{\Lambda})\big(-(2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0+2Cy_0\cos\lambda_0\big).$$

a,b,c - $x_0(\tau),y_0(\tau),\lambda_0(\tau),\Lambda_0(\tau)$:

$$a(\tau)=\frac{\left(e_J^2D^2-4CEe_J^2\right)\sin\lambda_0\cos\lambda_0}{\beta(x_0,y_0)},\tag{23}$$

$$b(\tau)=-2F-2C\cos\lambda_0-\frac{1}{\beta(x_0,y_0)}\big((2Cy_0)\sin\lambda_0+(2Cx_0+2C\hat{x}_0+e_JD)\cos\lambda_0\big)^2,\tag{24}$$

$$c(\tau)=2F-2C\cos\lambda_0+\frac{1}{\beta(x_0,y_0)}\big((2Cx_0+2C\hat{x}_0+e_JD)\sin\lambda_0-(2Cy_0)\cos\lambda_0\big)^2.\tag{25}$$

$G_k^{\rho}, \rho\in\{\lambda,\Lambda,x,y\}$, (22), :

$$G_k^{\rho}=\sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^{\rho}}{j!},\quad k\geq 2,$$

g^{ρ} - , $\rho\in\{\lambda,\Lambda,x,y\}$,

$$\mathcal{L}_k^{(j)}=\sum_{k_1+\ldots+k_j=k}\mathfrak{D}_{k_1}\cdot\ldots\cdot\mathfrak{D}_{k_j},\qquad \mathfrak{D}_k=\left(\begin{pmatrix}\lambda_k\\ \Lambda_k\\ x_k\\ y_k\end{pmatrix},\begin{pmatrix}\frac{\partial}{\partial\lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial\Lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial x}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\\ \frac{\partial}{\partial y}\big|_{\lambda_0,\Lambda_0,x_0,y_0}\end{pmatrix}\right),$$

$(\cdot,\cdot)$ – $\mathbb{R}^4$.
:

$$\mathcal{F}_s = \left\{ f\colon \mathbb{R}\to\mathbb{R} \;\middle|\; \exists C>0:\ \forall x\in\mathbb{R} \Rightarrow\ |f(x)|\le Ce^{-s|x|} \right\}, s>0$$

$$\mathcal{F}_{-s} = \left\{ f\colon \mathbb{R}\to\mathbb{R} \;\middle|\; \exists C>0:\ \forall x\in\mathbb{R} \Rightarrow\ |f(x)|\ge Ce^{s|x|} \right\}, s>0$$

$\lambda_j, \Lambda_j, x_j, y_j \in \mathcal{F}_s \quad \forall j: \quad 0\le j\le k,\ \ g^\rho \in C_b^\infty(\mathbb{R}^4), \quad \forall \rho\in\{\Lambda,\lambda,x,y\}$ (,), :

$$G^\rho_{k+1} \in \mathcal{F}_{2s} \quad \forall \rho\in\{\Lambda,\lambda,x,y\}$$

, :

$$G^\lambda_k = 0 \quad \forall k\ge 1.$$

$\varepsilon = 0.$
, $W^{s,u}(0)$, :

$$\begin{cases} \Lambda_0(\tau)=0, \quad \lambda_0(\tau)=\lambda_-(x_{sep}(\tau),y_{sep}(\tau)), \\ x_0(\tau)=x_{sep}(\tau)-\hat{x}_0, \quad y_0(\tau)=y_{sep}(\tau). \end{cases}$$

(x_1,y_1) :

$$\frac{d}{d\tau}\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}. \tag{26}$$

. ( ). $\tilde{G}^x_1$ $\tilde{G}^y_1$ - .
(26) :

$$x_1^{s,u}(\tau) = \frac{dx_0}{d\tau}, \qquad y_1^{s,u}(\tau) = \frac{dy_0}{d\tau}, \qquad \boldsymbol{u}_1(\tau) \equiv \begin{pmatrix} x_1(\tau) \\ y_1(\tau) \end{pmatrix} \tag{27}$$

(26) ( ) $\tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) \equiv (\tilde{x}_1(\tau),\tilde{y}_1(\tau))$. .. $\mathcal{A}$ 0, , -, , , :

$$W(\tau) = x_1(\tau)\tilde{y}_1(\tau) - y_1(\tau)\tilde{x}_1(\tau) = 1$$

(x_1,y_1) $(\tilde{x}_1,\tilde{y}_1)$.
$\boldsymbol{u}_1(\tau)$ $\tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)$. $\mathcal{A}(\tau)$ $|\tau|$:

$$\mathcal{A}(\tau) = \begin{pmatrix} 0 & -2F-2C+\frac{(2C\hat{x}_0+e_JD)^2}{U_0} \\ 2F-2C & 0 \end{pmatrix} + \mathcal{O}(e^{-\xi_0^+|\tau|}) \equiv \begin{pmatrix} 0 & b_0 \\ c_0 & 0 \end{pmatrix} + \mathcal{O}(e^{-\xi_0^+|\tau|}), \quad \sqrt{b_0c_0} = \xi_0^+ > 0 \tag{28}$$

:

$$\boldsymbol{u}_1(\tau) = \mathrm{const}\begin{pmatrix} b_0\sqrt{c_0} \\ c_0\sqrt{b_0} \end{pmatrix} e^{-\xi_0^+|\tau|} + \mathcal{O}(e^{-2\xi_0^+|\tau|}) \in \mathcal{F}_{\xi_0^+}, \qquad \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau) = \mathrm{const}\begin{pmatrix} b_0\sqrt{c_0} \\ -c_0\sqrt{b_0} \end{pmatrix} e^{\xi_0^+|\tau|} + \mathcal{O}(1) \in \mathcal{F}_{-\xi_0^+}.$$

, $\boldsymbol{u}_1(\tau)$ $W^s(0)$, $W^u(0)$, s,u .
$\mathcal{A}(\tau)$, . , , :

$$\begin{cases} \boldsymbol{u}^u_k = \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau^u_k}^{\tau}\Big(\tilde{y}_1(\tau)\tilde{G}^x_k(\tau)-\tilde{x}_1(\tau)\tilde{G}^y_k(\tau)\Big)\,d\tau + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\displaystyle\int_{-\infty}^{\tau}\Big(x_1(\tau)\tilde{G}^y_k(\tau)-y_1(\tau)\tilde{G}^x_k(\tau)\Big)\,d\tau, \\ \boldsymbol{u}^s_k = \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau^s_k}^{\tau}\Big(\tilde{y}_1(\tau)\tilde{G}^x_k(\tau)-\tilde{x}_1(\tau)\tilde{G}^y_k(\tau)\Big)\,d\tau - \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\displaystyle\int_{\tau}^{+\infty}\Big(x_1(\tau)\tilde{G}^y_k(\tau)-y_1(\tau)\tilde{G}^x_k(\tau)\Big)\,d\tau. \end{cases} \tag{29}$$

, (29) $\tau \to \pm\infty$. , (29) τ .
$\mathbf{u}_1(\tau)$ $\tau \to \pm\infty$, $\mathbf{u}_k^u$ $\mathbf{u}_k^s$ $\tau_k^u, \tau_k^s \in \mathbb{R}$.
$\tilde{\mathbf{u}}_1(\tau)$ $\tau \to \pm\infty$. $\mathbf{u}_k^u$ $\mathbf{u}_k^s$, (29), $\tilde{\mathbf{u}}_1(\tau)$. $(x_1\tilde{G}_k^y - y_1\tilde{G}_k^x) \in \mathcal{F}_{3\xi_0^+}$, , , $\tilde{\mathbf{u}}_1$.
(20), (x_1, y_1) (22) $(x_1(\tau), y_1(\tau)) = \mathrm{const} \cdot \boldsymbol{u}_1$, τ_k^s .
- , $W^s(0)$. (18), $W^s(0)$.
$W^u(0)$, , $+\infty$.
:

$$|\boldsymbol{u}_k^u(\tau) - \boldsymbol{u}_k^s(\tau)| = \Bigg|\boldsymbol{u}_1(\tau)\int_{\tau_k^u}^{\tau_k^s}\Big(\tilde{y}_1(\tau)\tilde{G}_k^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_k^y(\tau)\Big)\,d\tau + \\ + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\int_{-\infty}^{+\infty}\Big(x_1(\tau)\tilde{G}_k^y(\tau) - y_1(\tau)\tilde{G}_k^x(\tau)\Big)\,d\tau\Bigg| \quad (30)$$

τ . $W^s(0)$ $W^u(0)$ $\boldsymbol{u}_k^s$ $\boldsymbol{u}_k^u$ τ.
, $|\boldsymbol{u}_k^u(\tau_1) - \boldsymbol{u}_k^s(\tau_2)| = 0$ τ_1, τ_2. , , $\tau_1 = \tau_2 = \tau$.

1. , $f(\tau)$, $f(\tau)$, τ.

, $\lambda_j, \Lambda_j, x_j, y_j$, $j \le k-1$ τ . (18) , $G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau), G_k^y(\tau)$. $G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau)$, $G_k^y(\tau), \tilde{G}_k^y(\tau)$ $G_k^x(\tau)$ ( ).
(18) $G_2^x(\tau), \tilde{G}_2^x(\tau), G_2^\Lambda$ - , $G_2^y(\tau), \tilde{G}_2^y(\tau)$ - .
, , :

$$\int_{-\infty}^{+\infty}\Big(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau)\Big)\,d\tau = 0. \quad (31)$$

4.1.2.

, 0.
, :

$$\boldsymbol{u}_1^0(\tau) \equiv \begin{pmatrix} x_1^0(\tau) \\ y_1^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$\tilde{G}_2^\rho, \rho \in \{\Lambda, \lambda, x, y\}$.
$\tau_2^s = \tau_2^u = 0$:

$$\begin{cases} \boldsymbol{u}_2^u = \boldsymbol{u}_1(\tau)\displaystyle\int_0^\tau\Big(\tilde{y}_1(\tau)\tilde{G}_2^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_2^y(\tau)\Big)\,d\tau + \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\displaystyle\int_{-\infty}^\tau\Big(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau)\Big)\,d\tau, \\ \boldsymbol{u}_2^s = \boldsymbol{u}_1(\tau)\displaystyle\int_0^\tau\Big(\tilde{y}_1(\tau)\tilde{G}_2^x(\tau) - \tilde{x}_1(\tau)\tilde{G}_2^y(\tau)\Big)\,d\tau - \tilde{\boldsymbol{u}}_{\mathbf{1}}(\tau)\displaystyle\int_\tau^{+\infty}\Big(x_1(\tau)\tilde{G}_2^y(\tau) - y_1(\tau)\tilde{G}_2^x(\tau)\Big)\,d\tau, \end{cases}$$

, (31)

$$\boldsymbol{u}_2^s(\tau) = \boldsymbol{u}_2^u(\tau) \equiv \boldsymbol{u}_2^0(\tau).$$

$\tilde{G}_k^\rho, \rho \in \{\Lambda, \lambda, x, y\}$. 3 . :

$$\boldsymbol{u}_3^0(\tau) \equiv \begin{pmatrix} x_3^0(\tau) \\ y_3^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad (\lambda_3^0(\tau), \Lambda_3^0(\tau)) = (0, 0).$$

, $k \ge 0$:

$$G^{\rho}_{2k+1} = 0, \rho \in \{\Lambda, \lambda, x, y\}, \qquad G^{x}_{2k+2}, y_{2k+2} - , \qquad G^{y}_{2k+2}, x_{2k+2} - .$$

:

$$(\lambda^0_{2k+1}, \Lambda^0_{2k+1}, x^0_{2k+1}, y^0_{2k+1}) = (0, 0, 0, 0).$$

:

$$\begin{cases} \lambda^0(\tau) = (\lambda_0(\tau) - \pi) + \sum_{k=1}^{+\infty} \varepsilon^{2k} \lambda^0_{2k}(\tau), & \Lambda^0(\tau) = \frac{1}{\alpha} \sum_{k=1}^{+\infty} \varepsilon^{2k} (\lambda^0_{2k-1})'(\tau), \\ x^0(\tau) = x_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} x^0_{2k}(\tau), & y^0(\tau) = y_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} y^0_{2k}(\tau), \end{cases}$$

$$\begin{cases} x^0_{2k}(\tau) = x_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau + \tilde{x}_1(\tau) \int_{-\infty}^\tau \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau, \\ y^0_{2k}(\tau) = y_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau + \tilde{y}_1(\tau) \int_{-\infty}^\tau \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau, \end{cases}$$

$$\tilde{G}^x_{2k} = \frac{1}{\alpha\beta} (\lambda^0_{2k-2})''(\tau) \left(-2Cy_0 \sin\lambda_0 - (2Cx_0 + 2C\hat{x}_0 + e_J D) \cos\lambda_0 \right),$$

$$\tilde{G}^y_{2k} = \frac{1}{\alpha\beta} (\lambda^0_{2k-2})''(\tau) \left(-(2Cx_0 + 2C\hat{x}_0 + e_J D) \sin\lambda_0 + 2Cy_0 \cos\lambda_0 \right).$$

1. (18):

- $(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) \in W^s(0)$,
- $(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) \in W^u(0)$,

ε $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$. $\exists$ $c > 0$:

$$(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}),$$

$$(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}).$$

:

$$\int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}^x_{2k}(\tau) - \tilde{x}_1(\tau) \tilde{G}^y_{2k}(\tau) \right) d\tau,$$

$\forall k \in \mathbb{N}$, .. $\mathcal{F}_{s_0}$.
(2) 0:

$$\int_{-\infty}^{+\infty} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau = 0, \qquad k \in \mathbb{N}$$

, $\mathcal{F}_{3s_0}$,

$$\int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}^y_{2k}(\tau) - y_1(\tau) \tilde{G}^x_{2k}(\tau) \right) d\tau$$

$\mathcal{F}_{2\xi_0^+}$. $\tilde{\boldsymbol{u}}_{\mathbf{1}}$.
, $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$ $\tau \to \pm\infty$.
, $(g^\Lambda, g^\lambda, g^x, g^y)$ (18) (Λ, λ, x, y) $\varepsilon > 0$. , $(g^\Lambda, g^\lambda, g^x, g^y)$ (Λ, λ, x, y) . .
, $W^s(0)$ $W^u(0)$ $(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$. .
■

1. - (18) .

4.2.

[6], - . [7] [8, 9]. .

4.2.1.

(18) . :

$$A = \begin{pmatrix} 0 & -U_0 & 0 & (2C\hat{x}_0 + e_J D) \\ \alpha & 0 & 0 & 0 \\ 0 & \varepsilon(2C\hat{x}_0 + e_J D) & 0 & -\varepsilon(2F + 2C) \\ 0 & 0 & \varepsilon(2F - 2C) & 0 \end{pmatrix}.$$

, A $\pm i\omega, \pm\xi$ $(z, \eta, a, b)^T = R^{-1}(\Lambda, \lambda, x, y)^T$, $R-$. :

$$\frac{d}{dt}(z, \eta, a, b)^T = \mathrm{diag}(i\omega, -i\omega, \xi, -\xi)(z, \eta, a, b)^T + \boldsymbol{\mathcal{G}}(z, \eta, a, b), \tag{32}$$

$\boldsymbol{\mathcal{G}}$.

$\varphi(t), \psi(t)$, , [6]. :

$$\varphi(t) = \varphi_0 e^{i\theta(\varphi_0,\psi_0)t}, \qquad \psi(t) = \varphi_0 e^{-i\theta(\varphi_0,\psi_0)t}.$$

$\theta(\varphi_0, \psi_0)$ φ_0, ψ_0:

$$\theta(\varphi_0, \psi_0) = \omega + \sum_{k=1}^{+\infty} \theta_k \cdot (\varphi_0\psi_0)^k.$$

:

$$\begin{cases} z(\varphi, \psi) = \varphi + \sum_{k,j=1}^{\infty} \{z\}_{k,j} \varphi^k \psi^j, & \eta(\varphi, \psi) = \psi + \sum_{k,j=1}^{\infty} \{\eta\}_{k,j} \varphi^k \psi^j, \\ a(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{a\}_{k,j} \varphi^k \psi^j, & b(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{b\}_{k,j} \varphi^k \psi^j. \end{cases} \tag{33}$$

, z $(\varphi\psi)^k\varphi$, η $(\varphi\psi)^k\psi$. φ, ψ [6].
(32) :

$$\begin{cases} ((p-q)i\omega - i\omega)\{z\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{z\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_z(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + i\omega)\{\eta\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{\eta\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_\eta(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega - \xi)\{a\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{a\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_a(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + \xi)\{b\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{b\}_{p-r,q-r} = \{\boldsymbol{\mathcal{G}}_b(z, \eta, a, b)\}_{p,q}. \end{cases} \tag{34}$$

$\{\boldsymbol{\mathcal{G}}_\rho(z, \eta, a, b)\}_{p,q}, \rho \in \{z, \eta, a, b\}$ $\varphi^p\psi^q$ $\boldsymbol{\mathcal{G}}$.
$p = q + 1, p + q > 1$ z , $\{z\}_{p,q} = 0$. $q = p + 1$ η , $\{\eta\}_{p,q} = 0$ ($(\varphi\psi)^k\varphi$ z $(\varphi\psi)^k\psi$ η). $\{z\}_{1,0} = \{\eta\}_{0,1} = 1$. ,<

$$\theta_p = \{\boldsymbol{\mathcal{G}}_z\}_{p,p-1} = -\{\boldsymbol{\mathcal{G}}_z\}_{p-1,p} \quad p > 1$$

(Λ, λ, x, y), , : $\varphi_0 = \psi_0 \equiv h$.

4.2.2.

.

2. , $f(x)$, - 0,

$$g(s) = \sum_{n=0}^{+\infty} g_n s^n, \quad s \geq 0, \quad g_n \geq 0 \quad \forall n \geq 0,$$

:

$f(x)$:

$$f(x) = \sum_{n=0}^{+\infty} f_n x^n.$$

$$\exists C > 0 : |f_n| \leq C \cdot g_n \quad \forall n \geq 0.$$

$$f \prec g.$$

$\varkappa$:

$$\varkappa = \max\left(\sup_{p,q\in\mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - i\omega} \right| \right), \sup_{p,q\in\mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - i\omega} \right| \right) \right) =$$
$$= \max(\frac{1}{|\xi|}, \frac{1}{|\omega|}) = \frac{1}{|\xi|} = O\left(\frac{1}{\varepsilon}\right) \qquad \varepsilon.$$

, (34), :

$$|\{\rho\}_{p,q}| \leq \varkappa |\{\boldsymbol{\mathcal{G}}_\rho(z,\eta,a,b)\}_{p,q}| + \varkappa \sum_{r=1}^{+\infty} |(p-q)\theta_r \{\rho\}_{p-r,q-r}|, \qquad \rho \in \{z,\eta,a,b\} \tag{35}$$

[6] , $\boldsymbol{\mathcal{G}}$

$$\boldsymbol{\mathcal{G}} \prec \frac{1}{1-s}, \qquad s \equiv |z| + |\eta| + |a| + |b|.$$

$\boldsymbol{\mathcal{G}}$.

.

(Λ, λ, x, y) (z, η, a, b) R, s:

$$s = |z| + |\eta| + \varepsilon(|a| + |b|).$$

R :

$$\Lambda \prec s, \qquad \lambda \prec s, \qquad x \prec \varepsilon s, \qquad y \prec \varepsilon s. \tag{36}$$

(36) , 0 , (18):

$$\begin{cases} -U\sin\lambda + V\cos\lambda \prec (\varepsilon s + \varepsilon^2 s^2)e^s, & \alpha\Lambda \prec s, \\ -\varepsilon\big(2Fy - \frac{\partial U}{\partial y}\cos\lambda - \frac{\partial V}{\partial y}\sin\lambda\big) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2)e^s, & \varepsilon\big(2F(x+\hat{x}_0) + e_J G - \frac{\partial U}{\partial x}\cos\lambda - \frac{\partial V}{\partial x}\sin\lambda\big) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2)e^s. \end{cases}$$

$\boldsymbol{\mathcal{G}}$:

$$\begin{cases} \boldsymbol{\mathcal{G}}_{z,\eta} & \prec \varepsilon\left((s+\varepsilon s^2)e^s - s\right), \\ \boldsymbol{\mathcal{G}}_{a,b} & \prec \left((s+\varepsilon s^2)e^s - s\right). \end{cases} \tag{37}$$

, $-s$ $(s+\varepsilon s^2)e^s$ $(s+\varepsilon s^2)e^s - s$.

h, φ, ψ, :

$$||h|| \equiv \sum_{p,q} |\{h\}_{p,q}| \varphi^p \psi^q.$$

(35) ε a b, $\varphi^p\psi^q$. :

$$(\varphi+\psi)||\theta-\omega||+S \prec \varkappa\big(||\boldsymbol{\mathcal{G}}_z||+||\boldsymbol{\mathcal{G}}_\eta||+\varepsilon(||\boldsymbol{\mathcal{G}}_a||+||\boldsymbol{\mathcal{G}}_b||)+||\theta-\omega||S\big), \tag{38}$$

$S=||z-\varphi||+||\eta-\psi||+\varepsilon(||a||+||b||)$.
$\varphi=\psi>0$. :

$$2\varphi||\theta-\omega||+S \prec \varkappa\left(\varepsilon(s+\varepsilon s^2)e^s-\varepsilon s+||\theta-\omega||S\right),$$

$s \prec 2\varphi+S$.
$Y=2||\theta-\omega||+\frac{1}{\varphi}S$, :

$$Y \prec \frac{\varkappa}{4}\left(Y^2+\varepsilon\left((2+Y)+\varepsilon\varphi(2+Y)^2\right)e^{\varphi(2+Y)}-(2+Y)\right). \tag{39}$$

(39), , $R=\mathcal{O}(\varepsilon)$, φ.

4.3.

, ?? $\Gamma_h(t)=(\Lambda^*(h,t),\lambda^*(h,t),x^*(h,t),y^*(h,t))$, h (,), $|h|\le R(\varepsilon)$ $R(\varepsilon)=O(\varepsilon)$ (ε -).
$W^s(\Gamma_h)$ $W^u(\Gamma_h)$. , , , Γ_h :

$$\begin{cases}\Lambda^{s,u}(h,t)=\Lambda^*(h,t)+z_\Lambda^{s,u}(h,t),\\ \lambda^{s,u}(h,t)=\lambda^*(h,t)+z_\lambda^{s,u}(h,t),\\ x^{s,u}(h,t)=x^*(h,t)+z_x^{s,u}(h,t),\\ y^{s,u}(h,t)=y^*(h,t)+z_y^{s,u}(h,t),\end{cases} \tag{40}$$

s , u . $z_\Lambda^{s,u},z_\lambda^{s,u},z_x^{s,u},z_y^{s,u}$:

- $W^s(\Gamma_h)$:
$$z_\Lambda^s(t)\to 0,\ z_\lambda^s(t)\to 0,\ z_x^s(t)\to 0,\ z_y^s(t)\to 0 \qquad t\to+\infty;$$
- $W^u(\Gamma_h)$:
$$z_\Lambda^u(t)\to 0,\ z_\lambda^u(t)\to 0,\ z_x^u(t)\to 0,\ z_y^u(t)\to 0 \qquad t\to-\infty.$$

, , -.
:

$$\Gamma_h(t)=(\Lambda^*(h,t),\lambda^*(h,t),x^*(h,t),y^*(h,t)). \tag{41}$$

$W^s(\Gamma_h)$ $W^u(\Gamma_h)$. , , :

$$\begin{cases}\Lambda^{s,u}(h,t)=\Lambda^*(h,t)+z_\Lambda^{s,u}(h,t), & \lambda^{s,u}(h,t)=\lambda^*(h,t)+z_\lambda^{s,u}(h,t),\\ x^{s,u}(h,t)=x^*(h,t)+z_x^{s,u}(h,t), & y^{s,u}(h,t)=y^*(h,t)+z_y^{s,u}(h,t),\end{cases} \tag{42}$$

$z_\Lambda^{s,u},z_\lambda^{s,u},z_x^{s,u},z_y^{s,u}$:

- $W^s(\Gamma_h)$:
$$z_\Lambda^s(t)\to 0,\ z_\lambda^s(t)\to 0,\ z_x^s(t)\to 0,\ z_y^s(t)\to 0 \qquad t\to+\infty;$$
- $W^u(\Gamma_h)$:
$$z_\Lambda^u(t)\to 0,\ z_\lambda^u(t)\to 0,\ z_x^u(t)\to 0,\ z_y^u(t)\to 0 \qquad t\to-\infty.$$

$\Theta(h)$ h. $\tilde{t} = \frac{\theta(h)}{\omega} t = (1 + \mathcal{O}(h^2))t$. , h h ω, $\hat{H}\left(\boldsymbol{z}, \tilde{t}\right)$ $\boldsymbol{z}$, , .
$\omega = \mathcal{O}(1)$ $\hat{H}$ $\xi = \mathcal{O}(\varepsilon)$, ($\langle\cdot\rangle$), :

$$\langle \hat{H} \rangle = H(\boldsymbol{z}) - h^2\theta_1 H(\boldsymbol{z}) + \alpha\langle\Lambda^*\rangle z_\Lambda + 2F\langle x^*\rangle z_x - \\ - \langle\cos\lambda^* - 1\rangle\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) - \\ - \cos z_\lambda\Big((2C\hat{x}_0 + e_J D)\langle y^*\sin\lambda^*\rangle_0 + z_x 2C\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right) + \Big\langle\big(U(x^*, y^*) - U_0\big)\cos\lambda^*\Big\rangle\Big) - \\ - \sin z_\lambda\Big(z_y 2C\left(\langle x^*\cos\lambda^*\rangle + \langle y^*\sin\lambda^*\rangle\right)\Big) + \mathcal{O}(h^4). \quad (43)$$

h, $\langle\hat{H}\rangle$

$$(\hat{\Lambda}_0^*(h), \hat{\lambda}_0^*(h), \hat{x}_0^*(h), \hat{y}_0^*(h)), \quad (44)$$

$\mathcal{O}(h^2)$.
, , [10, 11], ( ). , h , .
(44) 0 $\mathcal{O}(h^4)$, :

$$\hat{H}_0 = \frac{\alpha\omega z_\Lambda^2}{2\theta(h)} - \hat{U}(z_x, z_y)\cos z_\lambda - \hat{V}(z_x, z_y)\sin z_\lambda + \frac{F\omega}{\theta(h)}\left((z_x + \hat{x}_0 + \hat{x}_0^*(h))^2 + z_y^2\right) + \\ + \frac{\omega}{\theta(h)}\left(e_J G + 2F\langle x^*\rangle_0 + 2F\hat{x}_0^*(h)\right) z_x, \quad (45)$$

$\hat{U}$ $\hat{V}$ U, V $\mathcal{O}(h^2)$. $[\cdot]_0$ $\mathcal{O}(h^4)$.
, h (19) (45) ($\mathcal{O}(h^2)$) 1, (45) , 0 - (45) , .

4.3.1.

(41) $\mathcal{O}(\varepsilon)$ (.. $|h| \le R(\varepsilon) = \mathcal{O}(\varepsilon)$), $\mu \equiv h/\varepsilon$:

$$|\mu| \le \frac{R(\varepsilon)}{\varepsilon} = \mathcal{O}(1)$$

(45) $H(\boldsymbol{z})$:

$$\hat{H}_0(\boldsymbol{z}) = H(\boldsymbol{z}) + \varepsilon^2\mu^2 H_1(\boldsymbol{z}), \quad (46)$$

:

$$H_1(\boldsymbol{z}) \equiv \Bigg[-\theta_1 H_{old}(\boldsymbol{z}) + 2F z_x \underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} - \underbrace{\frac{1}{4}\left(\frac{2(\xi^2 - \alpha U_0)}{(2C\hat{x}_0 + e_J D)}\right)^2}_{\mathcal{O}(1)}\Big(U(z_x, z_y)\cos z_\lambda + V(z_x, z_y)\sin z_\lambda\Big) - \\ - \cos z_\lambda\left(2Cz_x + (2C\hat{x}^0 + e_J D)\right)\underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)} - \sin z_\lambda 2Cz_y \underbrace{(-2\varepsilon\xi)\{a\}_{1,1}}_{\mathcal{O}(1)}\Bigg]_0.$$

(46) (20):

$$z_\Lambda^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty}\varepsilon^k z_{\Lambda,k}^{s,u}(\tau,\mu), \quad z_\lambda^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty}\varepsilon^k z_{\lambda,k}^{s,u}(\tau,\mu), \quad z_x^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty}\varepsilon^k z_{x,k}^{s,u}(\tau,\mu), \quad z_y^{s,u}(\tau,\mu) = \sum_{k=0}^{\infty}\varepsilon^k z_{y,k}^{s,u}(\tau,\mu), \quad (47)$$

:

- $W^s(0)$:

$$z^s_\Lambda(+\infty)=0,\quad z^s_\lambda(+\infty)=0,\qquad z^s_x(+\infty)=0,\qquad z^s_y(+\infty)=0,$$

- $W^u(0)$:

$$z^u_\Lambda(-\infty)=0,\qquad z^u_\lambda(-\infty)=0,\qquad z^u_x(-\infty)=0,\qquad z^u_y(-\infty)=0.$$

(47) (46):

$$\begin{aligned}
&\frac{dz_\Lambda}{dt}=-U\sin z_\lambda+V\cos z_\lambda-\varepsilon^2\mu^2\frac{\partial H_1}{\partial z_\lambda}+\mathcal{O}(\varepsilon^4), &&\frac{dz_x}{dt}=-\varepsilon\left(2Fz_y-\frac{\partial U}{\partial z_y}\cos z_\lambda-\frac{\partial V}{\partial z_y}\sin z_\lambda+\varepsilon^2\mu^2\frac{\partial H_1}{\partial z_y}+\mathcal{O}(\varepsilon^4)\right),\\
&\frac{dz_\lambda}{dt}=\alpha z_\Lambda+\varepsilon^2\mu^2\frac{\partial H_1}{\partial z_\Lambda}+\mathcal{O}(\varepsilon^4), &&\frac{dz_y}{dt}=\varepsilon\left(2F(z_x+\hat{x}_0)+e_JG-\frac{\partial U}{\partial z_x}\cos z_\lambda-\frac{\partial V}{\partial z_x}\sin z_\lambda+\varepsilon^2\mu^2\frac{\partial H_1}{\partial z_x}+\mathcal{O}(\varepsilon^4)\right).
\end{aligned}\tag{48}$$

ε.

. :

$$z_{\Lambda,0}(\tau)=0,\qquad z_{\lambda,0}(\tau)=\lambda_-(x_{sep}(\tau),y_{sep}(\tau)),\qquad z_{x,0}(\tau)=x_{sep}(\tau)-\hat{x}_0,\qquad z_{y,0}(\tau)=y_{sep}(\tau).$$

$(z_{\Lambda,1},z_{\lambda,1},z_{x,1},z_{y,1})$ $(z_{\Lambda,1},z_{\lambda,1})$:

$$\begin{cases}
z_{\Lambda,1}(\tau)=z'_{\lambda,0},\\
z_{\lambda,1}(\tau)=\frac{1}{\beta}\Big(z_{x,1}\big((2Cz_{x,0}+2C\hat{x}_0+e_JD)\sin z_{\lambda,0}-2Cz_{y,0}\cos z_{\lambda,0}\big)+\\
+z_{y,1}\big(-2Cz_{y,0}\sin z_{\lambda,0}-(2Cz_{x,0}+2C\hat{x}_0+e_JD)\cos z_{\lambda,0}\big)\Big),
\end{cases}$$

$(z_{x,1},z_{y,1})$:

$$\frac{d}{d\tau}\begin{pmatrix}z_{x,1}\\z_{y,1}\end{pmatrix}=\mathcal{A}(\tau)\begin{pmatrix}z_{x,1}\\z_{y,1}\end{pmatrix}$$

$\mathcal{A}(\tau)$ - (22). - $\boldsymbol{u}_1(\tau),\tilde{\boldsymbol{u}_1}(\tau)$.
$W^s(\Gamma_h)$ $W^u(\Gamma_h)$, . :

$$(z_{x,1}(\tau),z_{y,1}(\tau))\equiv(0,0)$$

, μ, μ .
$(z_{x,2},z_{y,2})$, μ:

$$\frac{d}{d\tau}\begin{pmatrix}z_{x,2}\\z_{y,2}\end{pmatrix}=\mathcal{A}(\tau)\begin{pmatrix}z_{x,2}\\z_{y,2}\end{pmatrix}+\begin{pmatrix}\tilde{G}^x_2(\tau)\\\tilde{G}^y_2(\tau)\end{pmatrix}+\mu^2\begin{pmatrix}\mathfrak{G}^x_2(\tau)\\\mathfrak{G}^y_2(\tau)\end{pmatrix},\tag{49}$$

:

$$\mathfrak{G}^x_2=\frac{\partial H_1}{\partial z_y}\Big|_{z_{\Lambda,0},z_{\lambda,0},z_{x,0},z_{y,0}}-$$
$$-\beta^{-1}\left(\frac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0},z_{\lambda,0},z_{x,0},z_{y,0}}\right)\big(-2Cz_{y,0}\sin z_{\lambda,0}-(2Cz_{x,0}+2C\hat{x}_0+e_JD)\cos z_{\lambda,0}\big),$$

$$\mathfrak{G}^y_2=\frac{\partial H_1}{\partial z_x}\Big|_{z_{\Lambda,0},z_{\lambda,0},z_{x,0},z_{y,0}}-$$
$$-\beta^{-1}\left(\frac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0},z_{\lambda,0},z_{x,0},z_{y,0}}\right)\big(-(2Cz_{x,0}+2C\hat{x}_0+e_JD)\sin z_{\lambda,0}+2Cz_{y,0}\cos z_{\lambda,0}\big).$$

$$\begin{cases} z_{\Lambda,2}(\tau) = \dfrac{1}{\alpha} z'_{\lambda,1}(\tau), \\ z_{\lambda,2}(\tau) = \dfrac{1}{\beta}\Big(\alpha^{-1} z''_{\lambda,0} + \dfrac{\partial H_1}{\partial z_\lambda}\Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} + \\ + z_{x,2}\big((2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} - 2Cz_{y,0}\cos z_{\lambda,0}\big) + \\ + z_{y,2}\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big)\Big). \end{cases}$$

, $\mathfrak{G}_2^x(\tau)$ – , $\mathfrak{G}_2^y(\tau)$ – .
, $\tilde{G}_2^x, \tilde{G}_2^y$ $\boldsymbol{z}_0, \boldsymbol{z}_1$:

$$\tilde{G}_2^x = -\beta^{-1}\left(\alpha^{-1} z''_{\lambda,0}\right)\big(-2Cz_{y,0}\sin z_{\lambda,0} - (2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\cos z_{\lambda,0}\big),$$

$$\tilde{G}_2^y = -\beta^{-1}\left(\alpha^{-1} z''_{\lambda,0}\right)\big(-(2Cz_{x,0} + 2C\hat{x}_0 + e_J D)\sin z_{\lambda,0} + 2Cz_{y,0}\cos z_{\lambda,0}\big).$$

, (49) (22) , (49) :

$$\begin{cases} z_{x,2}(\tau,\mu) = z^0_{x,2}(\tau) + \hat{z}_{x,2}(\tau,\mu), \\ z_{y,2}(\tau,\mu) = z^0_{y,2}(\tau) + \hat{z}_{y,2}(\tau,\mu), \end{cases}$$

$(z^0_{x,2}, z^0_{y,2})$ - (22), $(\hat{z}_{x,2}, \hat{z}_{y,2})$ - :

$$\frac{d}{d\tau}\begin{pmatrix}\hat{z}_{x,2}\\ \hat{z}_{y,2}\end{pmatrix} = \mathcal{A}(\tau)\begin{pmatrix}\hat{z}_{x,2}\\ \hat{z}_{y,2}\end{pmatrix} + \mu^2\begin{pmatrix}\mathfrak{G}_2^x(\tau)\\ \mathfrak{G}_2^y(\tau)\end{pmatrix},$$

$\boldsymbol{u}_2 = (x_2, y_2)$, $\boldsymbol{u}_2^0 = (x_2^0, y_2^0)$, $\hat{\boldsymbol{u}}_{\boldsymbol{2}} = (\hat{x}_2, \hat{y}_2)$. $\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s},\boldsymbol{u}}$ (s u):

$$\begin{cases} \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}} = \mu^2 \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau^u}^{\tau}\left(\tilde{y}_1(\tau)\mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau)\mathfrak{G}_2^y(\tau)\right)d\tau + \mu^2\tilde{\boldsymbol{u}}_{\boldsymbol{1}}(\tau)\displaystyle\int_{-\infty}^{\tau}\left(x_1(\tau)\mathfrak{G}_2^y(\tau) - y_1(\tau)\mathfrak{G}_2^x(\tau)\right)d\tau, \\ \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}} = \mu^2 \boldsymbol{u}_1(\tau)\displaystyle\int_{\tau^s}^{\tau}\left(\tilde{y}_1(\tau)\mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau)\mathfrak{G}_2^y(\tau)\right)d\tau - \mu^2\tilde{\boldsymbol{u}}_{\boldsymbol{1}}(\tau)\displaystyle\int_{\tau}^{+\infty}\left(x_1(\tau)\mathfrak{G}_2^y(\tau) - y_1(\tau)\mathfrak{G}_2^x(\tau)\right)d\tau. \end{cases}$$

, $W^u(\Gamma_{h_1})$ $W^s(\Gamma_{h_2})$,

$$\boldsymbol{u}_2^u(\tau_1,\mu_1) - \boldsymbol{u}_2^s(\tau_2,\mu_2) = \left(\boldsymbol{u}_2^{0,u}(\tau_1) - \boldsymbol{u}_2^{0,s}(\tau_2)\right) + \left(\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}}(\tau_1,\mu_1) - \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}}(\tau_2,\mu_2)\right). \tag{50}$$

$\hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{u}}, \hat{\boldsymbol{u}}_{\boldsymbol{2}}^{\boldsymbol{s}}$ (50) :

$$B(\tau_1,\tau_2)\cdot\begin{pmatrix}\mu_1^2\\ \mu_2^2\end{pmatrix} = \begin{pmatrix}f_1(\tau_1,\tau_2)\\ f_2(\tau_1,\tau_2)\end{pmatrix}, \tag{51}$$

$$B_{1,1} = x_1(\tau_1)\int_{\tau^u}^{\tau_1}\left(\tilde{y}_1\mathfrak{G}_2^x - \tilde{x}_1\mathfrak{G}_2^y\right)d\tau + \tilde{x}_1(\tau_1)\int_{-\infty}^{\tau_1}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau,$$

$$B_{1,2} = -x_1(\tau_2)\int_{\tau^s}^{\tau_2}\left(\tilde{y}_1\mathfrak{G}_2^x - \tilde{x}_1\mathfrak{G}_2^y\right)d\tau + \tilde{x}_1(\tau_2)\int_{\tau_2}^{+\infty}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau,$$

$$B_{2,1} = y_1(\tau_1)\int_{\tau^u}^{\tau_1}\left(\tilde{y}_1\mathfrak{G}_2^x - \tilde{x}_1\mathfrak{G}_2^y\right)d\tau + \tilde{y}_1(\tau_1)\int_{-\infty}^{\tau_1}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau,$$

$$B_{2,2} = -y_1(\tau_2)\int_{\tau^s}^{\tau_2}\left(\tilde{y}_1\mathfrak{G}_2^x - \tilde{x}_1\mathfrak{G}_2^y\right)d\tau + \tilde{y}_1(\tau_2)\int_{\tau_2}^{+\infty}\left(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x\right)d\tau,$$

$$f_1 = -x_2^{0,u}(\tau_1) + x_2^{0,s}(\tau_1) =$$
$$= -x_1(\tau_1)\int_{\tau_2^u}^{\tau_1}\left(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\right)d\tau - \tilde{x}_1(\tau_1)\int_{-\infty}^{\tau_1}\left(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\right)d\tau +$$
$$+ x_1(\tau_2)\int_{\tau_2^s}^{\tau_2}\left(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\right)d\tau - \tilde{x}_1(\tau_2)\int_{\tau_2}^{+\infty}\left(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\right)d\tau,$$

$$f_2 = -y_2^{0,u}(\tau_1) + y_2^{0,s}(\tau_1) =$$
$$= -y_1(\tau_1)\int_{\tau_2^u}^{\tau_1}\left(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\right)d\tau - \tilde{y}_1(\tau_1)\int_{-\infty}^{\tau_1}\left(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\right)d\tau +$$
$$+ y_1(\tau_2)\int_{\tau_2^s}^{\tau_2}\left(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\right)d\tau - \tilde{y}_1(\tau_2)\int_{\tau_2}^{+\infty}\left(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\right)d\tau.$$

, (51) $\mu_1^2, \mu_2^2,$ $\{\tau_1, \tau_2, \tau^s, \tau^u, \tau_2^s, \tau_2^u\}$.
(51) $\det B(\tau_1, \tau_2) \neq 0$.
$\tau^u = \tau_1,\ \tau^s = \tau_2,$:

$$\det B(\tau_1, \tau_2) = \int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau\,(\tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2) - \tilde{x}_1(\tau_2)\tilde{y}_1(\tau_1)) \equiv$$
$$\equiv F(\tau_1, \tau_2)\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau,$$

$$B_{1,1} = \tilde{x}_1(\tau_1)\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau,$$

$$B_{1,2} = \tilde{x}_1(\tau_2)\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau,$$

$$B_{2,1} = \tilde{y}_1(\tau_1)\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau,$$

$$B_{2,2} = \tilde{y}_1(\tau_2)\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau.$$

, $\mu_1, \mu_2 > 0,$ (51) $\mu_1^2 > 0, \mu_2^2 > 0.$:

$$\frac{B_{1,1}f_2 - B_{2,1}f_1}{\det B} > 0,$$
$$\frac{B_{2,2}f_1 - B_{1,2}f_2}{\det B} > 0,$$

$$B_{1,1}f_2 - B_{2,1}f_1 = \int_{-\infty}^{\tau_1} (x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau\Bigg($$
$$\int_{\tau_2^u}^{\tau_1} \Big(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\Big)\,d\tau\Big(\underbrace{x_1(\tau_1)\tilde{y}_1(\tau_1) - \tilde{x}_1(\tau_1)y_1(\tau_1)}_{W(\boldsymbol{u}_1,\tilde{\boldsymbol{u}}_1)=1}\Big)+$$
$$+\int_{\tau_2^s}^{\tau_2} \Big(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\Big)\,d\tau\Big(\tilde{x}_1(\tau_1)y_1(\tau_2) - \tilde{y}_1(\tau_1)x_1(\tau_2)\Big)+$$
$$+\int_{\tau_2}^{+\infty} \Big(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\Big)\,d\tau\Big(\underbrace{\tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2)}_{-F(\tau_1,\tau_2)}\Big)\Bigg),$$

$$B_{2,2}f_1 - B_{1,2}f_2 = \int_{\tau_2}^{+\infty} (x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau\Bigg($$
$$\int_{\tau_2^u}^{\tau_1} \Big(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\Big)\,d\tau\Big(\tilde{x}_1(\tau_2)y_1(\tau_1) - \tilde{y}_1(\tau_2)x_1(\tau_1)\Big)+$$
$$+\int_{\tau_2^s}^{\tau_2} \Big(\tilde{y}_1\tilde{G}_2^x - \tilde{x}_1\tilde{G}_2^y\Big)\,d\tau\Big(\underbrace{x_1(\tau_2)\tilde{y}_1(\tau_2) - \tilde{x}_1(\tau_2)y_1(\tau_2)}_{W(\boldsymbol{u}_1,\tilde{\boldsymbol{u}}_1)=1}\Big)+$$
$$+\int_{-\infty}^{\tau_1} \Big(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\Big)\,d\tau\Big(\underbrace{\tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2)}_{-F(\tau_1,\tau_2)}\Big)\Bigg).$$

, :

$$\frac{B_{1,1}f_2 - B_{2,1}f_1}{\det B} = -\frac{\int_{\tau_2}^{+\infty}\Big(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\Big)\,d\tau}{\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau} + \frac{\int_{\tau_2^u}^{\tau_1}\Big(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\Big)\,d\tau}{F(\tau_1,\tau_2)\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau}+$$
$$+\frac{\int_{\tau_2^s}^{\tau_2}\Big(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\Big)\,d\tau\,(\tilde{x}_1(\tau_1)y_1(\tau_2) - \tilde{y}_1(\tau_1)x_1(\tau_2))}{F(\tau_1,\tau_2)\int_{\tau_2}^{+\infty}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau} > 0,$$

$$\frac{B_{2,2}f_1 - B_{1,2}f_2}{\det B} = -\frac{\int_{-\infty}^{\tau_1}\Big(x_1\tilde{G}_2^y - y_1\tilde{G}_2^x\Big)\,d\tau}{\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau} + \frac{\int_{\tau_2^s}^{\tau_2}\Big(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\Big)\,d\tau}{F(\tau_1,\tau_2)\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau}+$$
$$+\frac{\int_{\tau_2^u}^{\tau_1}\Big(\tilde{x}_1\tilde{G}_2^y - \tilde{y}_1\tilde{G}_2^x\Big)\,d\tau\,(\tilde{x}_1(\tau_2)y_1(\tau_1) - \tilde{y}_1(\tau_2)x_1(\tau_1))}{F(\tau_1,\tau_2)\int_{-\infty}^{\tau_1}(x_1\mathfrak{G}_2^y - y_1\mathfrak{G}_2^x)\,d\tau} > 0,$$

$$F(\tau_1,\tau_2) \equiv \tilde{y}_1(\tau_1)\tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1)\tilde{y}_1(\tau_2).$$

5.

6.

6.1. 1

1) $g^\rho(\lambda, \Lambda, x, y)$ (20). :

$$g^\rho(\lambda, \Lambda, x, y) = g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \sum_{j=1}^{+\infty} \frac{\mathcal{T}^j g^\rho}{j!},$$

$\mathcal{T}$ - :

$$\mathcal{T} = \left(\sum_{k=1}^{+\infty} \lambda_k \varepsilon^k\right) \frac{\partial}{\partial \lambda}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} + \left(\sum_{k=1}^{+\infty} \Lambda_k \varepsilon^k\right) \frac{\partial}{\partial \Lambda}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} + \\ + \left(\sum_{k=1}^{+\infty} x_k \varepsilon^k\right) \frac{\partial}{\partial x}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0} + \left(\sum_{k=1}^{+\infty} y_k \varepsilon^k\right) \frac{\partial}{\partial y}\bigg|_{\lambda_0,\Lambda_0,x_0,y_0}.$$

$$\mathfrak{D}_k = \left(\begin{pmatrix} \lambda_k \\ \Lambda_k \\ x_k \\ y_k \end{pmatrix}, \begin{pmatrix} \frac{\partial}{\partial \lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0} \\ \frac{\partial}{\partial \Lambda}\big|_{\lambda_0,\Lambda_0,x_0,y_0} \\ \frac{\partial}{\partial x}\big|_{\lambda_0,\Lambda_0,x_0,y_0} \\ \frac{\partial}{\partial y}\big|_{\lambda_0,\Lambda_0,x_0,y_0} \end{pmatrix} \right),$$

:

$$\mathcal{T} = \sum_{k=1}^{+\infty} \varepsilon^k \mathfrak{D}_k.$$

$\mathcal{T}^j$, , :

$$\mathcal{T}^j = \left(\sum_{k=1}^{+\infty} \varepsilon^k \mathfrak{D}_k\right)^j = \sum_{k=j}^{+\infty} \varepsilon^k \mathcal{L}_k^{(j)},$$

:

$$\mathcal{L}_k^{(j)} = \sum_{k_1+\ldots+k_j=k} \mathfrak{D}_{k_1} \cdot \ldots \cdot \mathfrak{D}_{k_j}.$$

ε :

$$g^\rho(\lambda, \Lambda, x, y) = g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \sum_{k=1}^{+\infty} \varepsilon^k \left(\sum_{j=1}^{k} \frac{\mathcal{L}_k^{(j)} g^\rho}{j!}\right).$$

2 , $\mathcal{L}_k^{(1)} = \mathfrak{D}_k$:

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \varepsilon\mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k\left(\mathcal{L}_k^{(1)}g^\rho + \sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}\right) =$$

$$= g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \varepsilon\mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k\left(\mathfrak{D}_k g^\rho + \underbrace{\sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}}_{G_k^\rho}\right)$$

- $\mathfrak{D}_k g^\rho$ - $(\lambda_k,\Lambda_k,x_k,y_k)$;
- G_k^ρ - $(\lambda_j,\Lambda_j,x_j,y_j)$, $j \le k-1$ (22).

G_k^ρ $k \ge 2$, $k=1$ $G_1^\rho = 0$. :

$$G_k^\rho = \sum_{j=2}^{k}\frac{\mathcal{L}_k^{(j)}g^\rho}{j!}, \quad k \ge 2, \tag{52}$$

$$G_1^\rho \equiv 0,$$

$$g^\rho(\lambda,\Lambda,x,y) = g^\rho(\lambda_0,\Lambda_0,x_0,y_0) + \sum_{k=1}^{\infty}\varepsilon^k\mathfrak{D}_k g^\rho + \sum_{k=2}^{+\infty}\varepsilon^k G_k^\rho.$$

2) $g^\rho, \rho \in \{\lambda,\Lambda,x,y\}$, (52) $j=2$, G_{k+1}^ρ 2 $(\le k)$ $\mathfrak{D}_i\mathfrak{D}_j g^\rho, i+j \le k$. $\le k$ $\mathcal{F}_s$, , , G_{k+1}^ρ $\mathcal{F}_{2s}$.

3) $g^\lambda = \alpha\Lambda$, 0. $G_k^\lambda = 0$ $\forall k \ge 1$. ■

6.2.

1) 4 - :

$$\begin{cases}\dot\Lambda(t) = g^\Lambda(\Lambda,\lambda,x,y),\\ \dot\lambda(t) = g^\lambda(\Lambda,\lambda,x,y),\\ \dot x(t) = \varepsilon g^x(\Lambda,\lambda,x,y),\\ \dot y(t) = \varepsilon g^y(\Lambda,\lambda,x,y),\end{cases}$$

$\tau = \varepsilon t$:

$$\begin{cases}\varepsilon\Lambda'(\tau) = g^\Lambda(\Lambda,\lambda,x,y),\\ \varepsilon\lambda'(\tau) = g^\lambda(\Lambda,\lambda,x,y),\\ x'(\tau) = g^x(\Lambda,\lambda,x,y),\\ y'(\tau) = g^y(\Lambda,\lambda,x,y),\end{cases}$$

$\boldsymbol{X} = (\Lambda,\lambda,x,y)$: $\boldsymbol{X} = \sum_{k=0}^{\infty}\varepsilon^k\boldsymbol{X}_k$. ε :

$$\begin{cases}0 = g^\Lambda(\boldsymbol{X}_0),\\ 0 = g^\lambda(\boldsymbol{X}_0),\\ x_0'(\tau) = g^x(\boldsymbol{X}_0),\\ y_0'(\tau) = g^y(\boldsymbol{X}_0),\end{cases} \tag{53}$$

$$
\begin{cases}
\Lambda_0'(\tau) = (\nabla g^{\Lambda}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\
\lambda_0'(\tau) = (\nabla g^{\lambda}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\
x_1'(\tau) = (\nabla g^{x}(\boldsymbol{X}_0), \boldsymbol{X}_1), \\
y_1'(\tau) = (\nabla g^{y}(\boldsymbol{X}_0), \boldsymbol{X}_1),
\end{cases}
\tag{54}
$$

, (54) (Λ_1, λ_1), :

$$
\begin{pmatrix} (\nabla g^{\Lambda})_{\Lambda} & (\nabla g^{\Lambda})_{\lambda} \\ (\nabla g^{\lambda})_{\Lambda} & (\nabla g^{\lambda})_{\lambda} \end{pmatrix} \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = A \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} \Lambda_0' - (\nabla g^{\Lambda})_x x_1 - (\nabla g^{\Lambda})_y y_1 \\ \lambda_0' - (\nabla g^{\lambda})_x x_1 - (\nabla g^{\lambda})_y y_1 \end{pmatrix},
$$

$$
A = \begin{pmatrix} (\nabla g^{\Lambda})_{\Lambda} & (\nabla g^{\Lambda})_{\lambda} \\ (\nabla g^{\lambda})_{\Lambda} & (\nabla g^{\lambda})_{\lambda} \end{pmatrix}.
$$

:

$$
\begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = M \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \boldsymbol{C},
$$

$$
M = \frac{1}{\det A} \begin{pmatrix} -(\nabla g^{\lambda})_{\lambda}(\nabla g^{\Lambda})_x + (\nabla g^{\Lambda})_{\lambda}(\nabla g^{\lambda})_x & -(\nabla g^{\lambda})_{\lambda}(\nabla g^{\Lambda})_y + (\nabla g^{\Lambda})_{\lambda}(\nabla g^{\lambda})_y \\ (\nabla g^{\lambda})_{\Lambda}(\nabla g^{\Lambda})_x - (\nabla g^{\Lambda})_{\Lambda}(\nabla g^{\lambda})_x & (\nabla g^{\lambda})_{\Lambda}(\nabla g^{\Lambda})_y - (\nabla g^{\Lambda})_{\Lambda}(\nabla g^{\lambda})_y \end{pmatrix},
$$

$$
\boldsymbol{C} = \frac{1}{\det A} \begin{pmatrix} (\nabla g^{\lambda})_{\lambda}\Lambda_0' - (\nabla g^{\Lambda})_{\lambda}\lambda_0' \\ -(\nabla g^{\lambda})_{\Lambda}\Lambda_0' + (\nabla g^{\Lambda})_{\Lambda}\lambda_0' \end{pmatrix}.
$$

(54) :

$$
\begin{cases}
x_1'(\tau) = \Big((\nabla g^{x})_x + \big((\nabla g^{x})_{\Lambda} M_{11} + (\nabla g^{x})_{\lambda} M_{21}\big)\Big) x_1 + \\
\qquad + \Big((\nabla g^{x})_y + \big((\nabla g^{x})_{\Lambda} M_{12} + (\nabla g^{x})_{\lambda} M_{22}\big)\Big) y_1 + \\
\qquad + \left(\begin{pmatrix} (\nabla g^{x})_{\Lambda} \\ (\nabla g^{x})_{\lambda} \end{pmatrix}, \boldsymbol{C} \right), \\
\\
y_1'(\tau) = \Big((\nabla g^{y})_x + \big((\nabla g^{y})_{\Lambda} M_{11} + (\nabla g^{y})_{\lambda} M_{21}\big)\Big) x_1 + \\
\qquad + \Big((\nabla g^{x})_y + \big((\nabla g^{y})_{\Lambda} M_{12} + (\nabla g^{y})_{\lambda} M_{22}\big)\Big) y_1 + \\
\qquad + \left(\begin{pmatrix} (\nabla g^{y})_{\Lambda} \\ (\nabla g^{y})_{\lambda} \end{pmatrix}, \boldsymbol{C} \right).
\end{cases}
\tag{55}
$$

$(\cdot,\cdot)$ – $\mathbb{R}^n$.

, , , (18) . , :

$$
\Lambda_0 = 0 \quad \Rightarrow \quad \boldsymbol{C} = \begin{pmatrix} \frac{\lambda_0'}{\alpha} \\ 0 \end{pmatrix}
$$

, g^x, g^y Λ, , (55) :

$$\begin{pmatrix}(\nabla g^x)_\Lambda\\(\nabla g^x)_\lambda\end{pmatrix}=\begin{pmatrix}0*\end{pmatrix},$$
$$\begin{pmatrix}(\nabla g^y)_\Lambda\\(\nabla g^y)_\lambda\end{pmatrix}=\begin{pmatrix}0*\end{pmatrix}.$$

(55) :

$$\left(\begin{pmatrix}(\nabla g^x)_\Lambda\\(\nabla g^x)_\lambda\end{pmatrix},\boldsymbol{C}\right)=\left(\begin{pmatrix}(\nabla g^y)_\Lambda\\(\nabla g^y)_\lambda\end{pmatrix},\boldsymbol{C}\right)=0.$$

, (18) (x_1,y_1) .

2) (2.4). x_0.

x_0 , , Λ_0,λ_0 x_0,y_0:

$$\begin{aligned}x_0''=(\nabla g^x)_x x_0'+(\nabla g^x)_y y_0'+(\nabla g^x)_\Lambda\Lambda_0'+(\nabla g^x)_\lambda\lambda_0'=\\=\left((\nabla g^x)_x+(\nabla g^x)_\Lambda\frac{\partial\Lambda_0}{\partial x}+(\nabla g^x)_\lambda\frac{\partial\lambda_0}{\partial x}\right)x_0'+\\+\left((\nabla g^x)_y+(\nabla g^x)_\Lambda\frac{\partial\Lambda_0}{\partial y}+(\nabla g^x)_\lambda\frac{\partial\lambda_0}{\partial y}\right)y_0'.\end{aligned}\quad(56)$$

, $\Lambda_0(x_0,y_0),\lambda_0(x_0,y_0)$:

$$\begin{cases}0=g^\Lambda(\boldsymbol{X}_0),\\0=g^\lambda(\boldsymbol{X}_0).\end{cases}\quad(57)$$

x:

$$\begin{cases}(\nabla g^\Lambda)_x+(\nabla g^\Lambda)_y\underbrace{y_x}_{0}+(\nabla g^\Lambda)_\Lambda\cdot\Lambda_x+(\nabla g^\Lambda)_\lambda\cdot\lambda_x=0,\\(\nabla g^\lambda)_x+(\nabla g^\lambda)_y\underbrace{y_x}_{0}+(\nabla g^\lambda)_\Lambda\cdot\Lambda_x+(\nabla g^\lambda)_\lambda\cdot\lambda_x=0.\end{cases}$$

:

$$\begin{pmatrix}(\nabla g^\Lambda)_\Lambda&(\nabla g^\Lambda)_\lambda\\(\nabla g^\lambda)_\Lambda&(\nabla g^\lambda)_\lambda\end{pmatrix}\begin{pmatrix}\Lambda_x\\\lambda_x\end{pmatrix}=-\begin{pmatrix}(\nabla g^\Lambda)_x\\(\nabla g^\lambda)_x\end{pmatrix}.$$

(Λ_x,λ_x) :

$$\Lambda_x=\frac{(\nabla g^\Lambda)_\lambda(\nabla g^\lambda)_x-(\nabla g^\Lambda)_x(\nabla g^\lambda)_\lambda}{(\nabla g^\Lambda)_\Lambda(\nabla g^\lambda)_\lambda-(\nabla g^\Lambda)_\lambda(\nabla g^\lambda)_\Lambda},$$
$$\lambda_x=\frac{(\nabla g^\Lambda)_x(\nabla g^\lambda)_\Lambda-(\nabla g^\Lambda)_\Lambda(\nabla g^\lambda)_x}{(\nabla g^\Lambda)_\Lambda(\nabla g^\lambda)_\lambda-(\nabla g^\Lambda)_\lambda(\nabla g^\lambda)_\Lambda}.$$

, M:

$$\frac{\partial\Lambda_0}{\partial x}=M_{11},$$
$$\frac{\partial\lambda_0}{\partial x}=M_{21},$$

:

$$\frac{\partial\Lambda_0}{\partial y}=M_{12},$$

$$\frac{\partial \lambda_0}{\partial y} = M_{22}.$$

(56) :

$$(x_0')' = (\nabla g^x)_x x_0' + (\nabla g^x)_y y_0' + (\nabla g^x)_\Lambda \Lambda_0' + (\nabla g^x)_\lambda \lambda_0' = \\ = ((\nabla g^x)_x + (\nabla g^x)_\Lambda M_{11} + (\nabla g^x)_\lambda M_{21})\, x_0' + \\ + ((\nabla g^x)_y + (\nabla g^x)_\Lambda M_{12} + (\nabla g^x)_\lambda M_{22})\, y_0'. \quad (58)$$

, (18) (55) , (58) (55) .

, $\tau \to \pm\infty$:

$$\begin{cases} x_0(\tau) = \dfrac{\text{const}}{\cosh(f_\pm \cdot \tau)} + \mathcal{O}(e^{-2f_\pm|\tau|}), \quad \tau \to \pm\infty, \\ y_0(\tau) = \dfrac{\text{const} \cdot \sinh(f_\pm \cdot \tau)}{\cosh^2(f_\pm \cdot \tau)} + \mathcal{O}(e^{-2f_\pm|\tau|}), \quad \tau \to \pm\infty, \end{cases}$$

$f_\pm = \lim_{\tau\to\pm\infty} \frac{df^\pm(\tau)}{d\tau} = \frac{J^+\chi^+}{\sigma^+} \lim_{\tau\to\pm\infty} \frac{P^+(f^+(\tau))}{S^+(f^+(\tau))}$. $P^\pm$ $S^\pm$ $\cosh f^\pm$, :

$$f_+ = f_- = \frac{J^+\chi^+}{\sigma^+} \cdot \frac{1}{U_0} > 0$$

:

$$x_0, y_0 \in \mathcal{F}_{s_0}, \quad s_0 = f_+ = f_- > 0$$

:

$$\begin{cases} x_1(\pm\infty) = 0, \\ y_1(\pm\infty) = 0. \end{cases}$$

$$x_1, y_1 \in \mathcal{F}_{s_0}$$

, (x_1, y_1) $W^s(0)$, $W^u(0)$, s, u .

■

1. G_k^ρ :

$$G_{k+1}^\rho = \frac{1}{2}\sum_{m=1}^{k} \mathfrak{D}_m \mathfrak{D}_{k+1-m} g^\rho + \sum_{m=1}^{k} \mathfrak{D}_m \hat{G}_{k+1-m}^\rho,$$

$\hat{G}_{k+1}^\rho$ G_{k+1}^ρ :

$$G_k^\rho = \sum_{j=2}^{k} \frac{\mathcal{L}_k^{(j)} g^\rho}{j!},$$

$$\hat{G}_k^\rho = \sum_{j=2}^{k} \frac{\mathcal{L}_k^{(j)} g^\rho}{(j+1)!},$$

$$\rho \in \lambda, \Lambda, x, y.$$

:

$$\mathcal{L}^{(j)}_{k+1} g^\rho = \sum_{m=1}^{k+2-j} \mathfrak{D}_m \mathcal{L}^{(j-1)}_{k+1-m} g^\rho$$

$$G^\rho_{k+1} = \sum_{j=2}^{k+1} \frac{\mathcal{L}^{(j)}_{k+1} g^\rho}{j!} = \sum_{j=2}^{k+1} \sum_{m=1}^{k+2-j} \frac{\mathcal{L}^{(j-1)}_{k+1-m} g^\rho}{j!} = \sum_{m=1}^{k} \mathfrak{D}_m \sum_{j=2}^{k+2-m} \frac{\mathcal{L}^{(j-1)}_{k+1-m} g^\rho}{j!} =$$

$$= \sum_{m=1}^{k} \mathfrak{D}_m \sum_{l=1}^{k+1-m} \frac{\mathcal{L}^{(l)}_{k+1-m} g^\rho}{(l+1)!} = \sum_{m=1}^{k} \mathfrak{D}_m \left(\sum_{l=2}^{k+1-m} \frac{\mathcal{L}^{(l)}_{k+1-m} g^\rho}{(l+1)!} + \frac{1}{2} \mathcal{L}^{(1)}_{k+1-m} g^\rho \right) =$$

$$= \sum_{m=1}^{k} \mathfrak{D}_m \left(\hat{G}^\rho_{k+1-m} + \frac{1}{2} \mathfrak{D}_{k+1-m} g^\rho \right) \blacksquare$$

2. 1) $\lambda_j, \Lambda_j, x_j, y_j, \quad j \le k-1 \quad \tau$,
$G^x_k(\tau), \tilde{G}^x_k(\tau), G^\Lambda_k(\tau), G^y_k(\tau)$. $G^x_k(\tau), \tilde{G}^x_k(\tau), G^\Lambda_k(\tau)$, $G^y_k(\tau), \tilde{G}^y_k(\tau)$ $G^x_k(\tau)$.

2) $G^x_2(\tau), \tilde{G}^x_2(\tau), G^\Lambda_2$ - , $G^y_2(\tau), \tilde{G}^y_2(\tau)$ - .

:

1) . .. $x_0(\tau)$ – , $y_0(\tau), \lambda_0(\tau)$ – :

$$g^\Lambda|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = U(x_0, y_0)\sin\lambda_0 - V(x_0, y_0)\cos\lambda_0 \quad \text{- ,}$$

$$g^x|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = -\varepsilon \left(2Fy_0 - \frac{\partial U}{\partial y}\cos\lambda_0 - \frac{\partial V}{\partial y}\sin\lambda_0 \right) \quad \text{- ,}$$

$$g^y|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} = \varepsilon \left(2F(x_0 + \hat{x}_0) + e_J G - \frac{\partial U}{\partial x}\cos\lambda_0 - \frac{\partial V}{\partial x}\sin\lambda_0 \right) \quad \text{- .}$$

:

- (x_0) .
- $(y_0 \quad \lambda_0)$.

, $+1$ -1 - , :

$$\begin{aligned}
\left.\frac{\partial^{p+q+m} g^\Lambda}{\partial x^p \partial y^q \partial \lambda^m}\right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} &\Leftrightarrow \quad -(-1)^{q+m}, \\
\left.\frac{\partial^{p+q+m} g^x}{\partial x^p \partial y^q \partial \lambda^m}\right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} &\Leftrightarrow \quad -(-1)^{q+m}, \\
\left.\frac{\partial^{p+q+m} g^y}{\partial x^p \partial y^q \partial \lambda^m}\right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} &\Leftrightarrow \quad (-1)^{q+m}.
\end{aligned} \tag{59}$$

, $G^\Lambda_k(\tau), G^x_k(\tau), G^y_k(\tau)$:

$$G^\Lambda_k(\tau) = \sum_{p_1,q_1,m_1,\ldots} \left.\frac{\partial^{p+q+m} g^\Lambda}{\partial x^p \partial y^q \partial \lambda^m}\right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

$$G^x_k(\tau) = \sum_{p_1,q_1,m_1,\ldots} \left.\frac{\partial^{p+q+m} g^x}{\partial x^p \partial y^q \partial \lambda^m}\right|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

$$G_k^y(\tau) = \sum_{p_1,q_1,m_1,\ldots} \frac{\partial^{p+q+m} g^y}{\partial x^p \partial y^q \partial \lambda^m}\bigg|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \ldots),$$

, .

, (.. $\leq k-1$), $G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau)$. , g^Λ g^x , g^y , .
$\tilde{G}_k^x, \tilde{G}_k^y$ $G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau)$.

2) , $x_0(\tau), y_1(\tau), \tilde{x}_1, \lambda_1$ – , $y_0(\tau), x_1(\tau), \tilde{y}_1$ – .

$$G_2^\rho(\tau) = \sum_{p+q+m=2,\quad p,q,m\geq 0,\quad p,q,m\neq 2} \frac{\partial^2 g^\rho}{\partial x^p \partial y^q \partial \lambda^m}\bigg|_{\lambda_0(\tau),\Lambda_0(\tau),x_0(\tau),y_0(\tau)} x_1^p y_1^q \lambda_1^m \tag{60}$$

, $x_1^p y_1^q \lambda_1^m$ $(-1)^p$. , (59), (60) $-(-1)^{p+q+m}$ $\rho \in \{\Lambda, x\}$ $(-1)^{p+q+m}$ $\rho = y$. $p+q+m=2$, :
$G_2^x(\tau), G_2^\Lambda$ - , $G_2^y(\tau)$ - .
$\tilde{G}_2^x, \tilde{G}_2^y$ $G_2^\Lambda(\tau), G_2^x(\tau), G_2^y(\tau)$. ■

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